

Work Stream 2: Deriving the Thermal Energy Equation and Vaporization Limit

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1 Introduction

A central question raised by [u/vibe0009](#) in [r/FluidMechanics](#) was how enstrophy divergence implies thermal shock when the incompressible Navier-Stokes equations lack a temperature equation. To address this, we must step outside the idealized incompressible frame and couple the viscous dissipation to the thermal energy equation.

2 Thermal Coupling

The complete energy equation for a fluid includes the viscous dissipation function Φ :

$$\rho c_p \left(\frac{\partial T}{\partial t} + u \cdot \nabla T \right) = k \nabla^2 T + 2\mu \mathbf{D}(u) : \mathbf{D}(u) \quad (1)$$

The contraction of the strain rate tensor $\mathbf{D}(u)$ is directly proportional to the enstrophy density $|\nabla \times u|^2$ in incompressible flow.

3 Heating by Pure Force

[u/Little-Name9809](#) asked whether this suggests a mechanism for vaporizing a fluid using only mechanical force (no external heat). In the OpenAI construction the *local* enstrophy density scales as $|\omega|^2 \sim \tau^{-2-2h} = \tau^{-2.010}$ (the *integrated* enstrophy $\Omega = \int |\omega|^2 dx$, which also absorbs the shrinking volume, scales as $\tau^{-0.515}$; $h = 1/200$), so the local heat generation rate $\varepsilon = \nu |\omega|^2$ diverges.

The temperature rise follows in three lines. The core keeps a radial Reynolds number of order one, so its radius is the diffusive length $\ell_r^2 = \nu t$, where t is the physical time remaining. The dissipation per unit mass is $\varepsilon = \nu (u/\ell_r)^2$. Acting over the time t ,

$$\Delta T \simeq \frac{\varepsilon t}{c_p} = \frac{\nu u^2 t}{\ell_r^2 c_p} = \frac{u^2}{c_p} \sim \tau^{-1.010}, \quad (2)$$

an Eckert number of order one: the kinetic energy per unit mass of the core is converted to heat within the core on its own collapse time. For water ($c_p = 4184 \text{ J/kg/K}$) this gives $\Delta T \approx 48 \text{ K}$ at $\text{Ma} = 0.3$ and $\approx 540 \text{ K}$ at $\text{Ma} = 1$; boiling ($\Delta T = 73 \text{ K}$) is reached at $u \approx 553 \text{ m/s}$, $\text{Ma} \approx 0.37$, about 3–4.5 ps before the mathematical blow-up (an earlier version of this note gave 45.6 ps, from an illustrative constant that has since been removed). Heating is therefore an order-one effect at the same scale, $\ell_* = \nu/c$, at which compressibility and rarefaction appear; it is not an independent, earlier limit. A plasma ($\sim 10^4 \text{ K}$) would need $\text{Ma} \approx 4$, far outside the incompressible model, so “plasma by pure force” is not supported by this estimate.

Caveat. The estimate assumes the heat stays in the core over the collapse time. The thermal diffusion length $\sqrt{\alpha t}$ compares with $\ell_r = \sqrt{\nu t}$ as $\text{Pr}^{-1/2}$: for water ($\text{Pr} \approx 7$) conduction removes only a fraction of the heat, so the estimate stands; for gases ($\text{Pr} \approx 0.7$) it is an upper bound. Either way the conclusion is that the constant-temperature, constant-density assumptions of the model stop holding at ℓ_* , not that any particular phase change is predicted.

Update (17 September 2026, v5.5.0): the estimate measured. A compressible Navier–Stokes–Fourier simulation of a forced collapsing core (`experiments/compressible_core.py`; paper §9.3) gives the actual temperature rise in a gas: the coefficient in $\Delta T = C u^2/c_p$ is $C \approx 0.35$ at $\text{Ma} \approx 0.4$, rising to $C \approx 1$ near $\text{Ma} \approx 1$, so u^2/c_p is an upper-order estimate, as the caveat anticipated. Two further findings: heat acts on the collapse only through the temperature dependence of viscosity; and on a Reynolds-number-one core (the route of the OpenAI construction) it does not stop the collapse before ℓ_* , whereas on a high-Reynolds-number (inertial) route air does not follow a driven swirl past a local Mach number of about 0.70 — a lock on the Mach number, not on velocity or core size, in a one-dimensional ideal-gas model.

4 Acknowledgments

Thank you to `u/vibe0009` and `u/Little-Name9809` for prompting this explicit thermal derivation. *Rendons à César ce qui appartient à César.*